

FEDERICO COLOMBO

Technical Artist | Unity Expert | Game Developer

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PROFESSIONAL PROFILE

Game & XR Developer actively pursuing a Master's Degree in Cinema and Media Engineering. Specialized in VR/XR development, Computer Graphics, and Gameplay Programming. Adept at bridging the gap between artistic vision and technical implementation using rendering pipelines and procedural generation to elevate interactive productions.

EDUCATION

M.Sc. Cinema and Media Engineering - Politecnico di Torino **2024 - 2026**

- **Current GPA / Expected Mark:** 106/110

B.Sc. Cinema and Media Engineering - Politecnico di Torino **2021 - 2024**

- **Final Mark:** 100/110

WORK EXPERIENCE

Lemons In The Room | Florence, Italy **03/2026 - Present**

XR Master Thesis - R&D Graphics Programmer

- Developed a 'context-aware' system for Mixed Reality that makes scanned real-world objects interactive and physically plausible.
- Implemented a real-time pipeline for Meta Quest 3 capable of converting Depth Maps into continuous geometry through Compute and Mesh Shaders, optimizing real-time performance.
- Performed GPU-based topological analysis fused with semantic data to generate procedural PBR materials dynamically.

Level Up Lab | Turin, Italy **01/2025 - Present**

Technical Artist & Gameplay Programmer

- Contributed to the design and implementation of core gameplay systems in Unity for a student team at Politecnico di Torino.
- Managed the creation and integration of all 3D models for the "Line Operator" project, overseeing artistic vision and technical viability.
- Developed custom Shaders and Post-Processing effects to establish and enhance the game's

unique visual identity.

Politecnico di Torino | Turin, Italy**10/2024 - 01/2025***Teaching Assistant (GUI)*

- Provided technical support and mentoring to students during lectures and laboratory sessions for the Graphical User Interfaces course.
- Created high-quality instructional content to improve students' comprehension of UI/UX principles and programming fundamentals.

RKH SRL | Turin, Italy**09/2023 - 06/2024***Video Artist*

- Operated professional cameras (RED, Blackmagic 4K/6K, Panasonic S1) and assisted the DOP with complex lighting setups.
- Handled editing and post-production workflows using DaVinci Resolve for music videos, podcasts, and corporate content.

SKILLS

- **Languages:** C#, C++, Python, HLSL/GLSL
- **Engines:** Unity, Godot, Unreal Engine
- **Software:** Blender, DaVinci Resolve, Jira, Git
- **Management:** Agile Methodology, Tech Direction, Project Management

PERSONAL PROJECTS

- **Awaken Asylum:** Immersive VR experience developed in Unity exploring the history of psychiatric institutions in Piedmont through meticulous historical reconstruction.
- **Nihil:** 2D video game revolving around a key transformation mechanic allowing players to change shape and unlock new abilities, programmed utilizing robust design patterns.